Definition 0.0.1. Let E be an Extension of field F.
Two elements $\alpha, \beta \in E$ which are algebraic over F are called *conjugate* over F if:

$$\operatorname{irr}(\alpha, F) = \operatorname{irr}(\beta, F)$$

where $\operatorname{irr}(\alpha, F), \operatorname{irr}(\beta, F) \in F[x]$ are monic irreducible polynomial having α, β as a root, respectively.

Theorem 0.0.2. Let E be an extension of a field F, and $\alpha, \beta \in E$ be algebraic over F, with $\deg(\alpha, F) = n$.
α and β are conjugate over F if and only if a conjugation map

$$\psi_{\alpha,\beta} : F(\alpha) \to F(\beta) : \sum_{i=0}^{n-1} c_i \cdot \alpha^i \mapsto \sum_{i=0}^{n-1} c_i \cdot \beta^i$$

is an isomorphism.

Proof. First, suppose that α, β are conjugate over F with respect to $p(x) \in F[x]$. Now, the evaluation homomorphisms

$$\phi_\alpha : F[x] \to F(\alpha) : f(x) \mapsto f(\alpha)$$
$$\phi_\beta : F[x] \to F(\beta) : f(x) \mapsto f(\beta)$$

have the same kernel $(p(x))$, because

$$f(x) \in (p(x)) \iff f(\alpha) = 0 \iff p(x) \mid f(x).$$

These induce natural field isomorphisms:

$$\varphi_\alpha : F[x]/(p(x)) \to F(\alpha) : f(x) + (p(x)) \mapsto f(\alpha)$$
$$\varphi_\beta : F[x]/(p(x)) \to F(\beta) : f(x) + (p(x)) \mapsto f(\beta)$$

Define an isomorphism $\psi_{\alpha,\beta}$ as the composition of the natural isomorphisms:

$$\psi_{\alpha,\beta} : F(\alpha) \to F(\beta) : x \mapsto (\varphi_\beta \circ \varphi_\alpha^{-1})(x)$$

This $\psi_{\alpha,\beta}$ maps $\sum_{i=0}^{n-1} c_i \alpha^i$ to $\sum_{i=0}^{n-1} c_i \beta^i$, because $\psi_{\alpha,\beta}(\alpha) = \varphi_\beta(\varphi_\alpha^{-1}(\alpha)) = \varphi_\beta(x) = \beta$ implies

$$\psi_{\alpha,\beta}\left(\sum_{i=0}^{n-1} c_i \alpha^i\right) = \sum_{i=0}^{n-1} c_i \cdot (\psi_{\alpha,\beta}(\alpha))^i = \sum_{i=0}^{n-1} c_i \beta^i.$$

Conversely, suppose that the given $\psi_{\alpha,\beta}$ is an isomorphism.
Put $\operatorname{irr}(\alpha, F), \operatorname{irr}(\beta, F) \in F[x]$ be monic irreducible polynomials of α, β, respectively. Then,

$$\psi_{\alpha,\beta}(\operatorname{irr}(\alpha, F)(\alpha)) = \operatorname{irr}(\alpha, F)(\beta) = 0$$
$$\psi_{\alpha,\beta}^{-1}(\operatorname{irr}(\beta, F)(\beta)) = \operatorname{irr}(\beta, F)(\alpha) = 0.$$

These imply $\operatorname{irr}(\alpha, F)$ divides $\operatorname{irr}(\beta, F)$ and $\operatorname{irr}(\beta, F)$ divides $\operatorname{irr}(\alpha, F)$. Since both are monic, they are the same. □

Corollary 0.0.3. Let $\alpha \in \overline{F}$ be algebraic over a field F.
For every homomorphism $\psi : F(\alpha) \to \overline{F}$ which fixes F, α and $\psi(\alpha)$ are conjugates.
Conversely, for each conjugate β of α over F, there exists one and only one homomorphism

$$\psi_{\alpha,\beta} : F(\alpha) \to \overline{F} : \sum_{i=0}^{\deg \alpha} c_i \alpha^i \mapsto \sum_{i=0}^{\deg \alpha} c_i \beta^i$$

which satisfies $\psi_{\alpha,\beta}(\alpha) = \beta$ and fixes F.

Proof. Suppose that $\psi : F(\alpha) \to \overline{F}$ is a homomorphism which fixes F.
Let $\operatorname{irr}(\alpha, F) = \sum_{i=0}^{n} a_i x^i \in F[x]$ be a monic minimal irreducible polynomial of α. Then,

$$\operatorname{irr}(\alpha, F)(\psi(\alpha)) = \sum_{i=0}^{n} a_i \psi(\alpha)^i = \psi\left(\sum_{i=0}^{n} a_i \alpha^i\right) = \psi(0) = 0.$$

That is, $\psi(\alpha)$ is a conjugate of α. □

0.1 Existence of Extension of Isomorphism

> **Theorem 0.1.1.** Let $\varphi : F \to F'$ be a field isomorphism.
> If E is an algebraic extension of F, then there exists an isomorphism $\phi : E \to E' \leq \overline{F'}$ s.t. $\phi|_F = \varphi$.

Proof. Define a set:

$$P = \left\{(L, \lambda) \mid L \text{ is field s.t. } F \leq L \leq E,\ \lambda : L \to \overline{F'} \text{ is field homomorphism s.t. } \lambda|_F = \varphi\right\}$$

Since $(F, \varphi) \in P$, P is non-empty. Define order on P as:

$$(L_1, \lambda_1) \leq (L_2, \lambda_2) \iff L_1 \leq L_2 \text{ and } \lambda_2|_{L_1} = \lambda_1$$

Let $T \overset{\text{def}}{=} \{(H_i, \lambda_i) \mid i \in I\}$ be a chain in P. Put $H \overset{\text{def}}{=} \bigcup_{i \in I} H_i$.
To show that H is a subfield of E, let $x, y \in H$ be given.
By definition, $x \in H_{i_1}$ and $y \in H_{i_2}$ for some $i_1, i_2 \in I$, and since T is a chain, either $H_{i_1} \subseteq H_{i_2}$ or $H_{i_2} \subseteq H_{i_1}$.
WLOG, assume $H_{i_1} \subseteq H_{i_2}$. Then, $x, y \in H_{i_2}$. Thus, we have

$$x \pm y,\ x \cdot y \in H_{i_2} \subseteq H \text{ and if } y \neq 0,\ \frac{x}{y} \in H_{i_2} \subseteq H$$

Therefore, H is a subfield of E containing F.
Meanwhile, define a map:

$$\lambda : H \to \overline{F'} : x \mapsto \lambda_i(x) \text{ if } x \in H_i \text{ for some } i \in I.$$

It is well-defined because: for any $x \in H$, $x \in H_i$ for some $i \in I$ by definition.
And, if $x \in H_i$ and H_j where $H_i \leq H_j$, then

$$\lambda_j(x) = \lambda_j|_{H_i}(x) = \lambda_i(x)$$

Thus $\lambda(x)$ is uniquely determined. And, it is a homomorphism because:
Let $x, y \in H$ be given. Then, for some $j \in I$, $x, y \in H_j$. Now,

$$\lambda(x + y) = \lambda_j(x + y) = \lambda_j(x) + \lambda_j(y) = \lambda(x) + \lambda(y)$$
$$\lambda(xy) = \lambda_j(xy) = \lambda_j(x)\lambda_j(y) = \lambda(x)\lambda(y)$$

And λ must fix F, thus non-trivial homomorphism. Finally, (H, λ) is an upper bound of T in P.
By Zorn's Lemma, there exists a maximal element $(K, \Phi) \in P$. To show that $K = E$, suppose that $K \neq E$. Choose $\alpha \in E \setminus K$.
Since $\alpha \in E$ is algebraic over F, it is also algebraic over K.
Let $p(x) \in K[x]$ be a monic irreducible polynomial such that $p(\alpha) = 0$.
Denote $K' = \Phi[K] \leq \overline{F'}$ and

$$p(x) = a_0 + a_1 x + \cdots + a_n x^n.$$

Consider the canonical isomorphism

$$\psi_\alpha : K[x]/(p(x)) \to K(\alpha) : f(x) + (p(x)) \mapsto f(\alpha)$$

And

$$\overline{p}(x) = \Phi(a_0) + \Phi(a_1)x + \cdots + \Phi(a_n)x^n \in K'[x]$$

Since Φ is an isomorphism, $\overline{p}$ is irreducible in K'.
Moreover, for root $\alpha' \in \overline{F'}$ of $\overline{p}(x)$, let

$$\psi_{\alpha'} : K'[x]/(\overline{p}(x)) \to K'(\alpha') : f(x) + (\overline{p}(x)) \mapsto f(\alpha')$$

is a canonical isomorphism.
Let

$$\overline{\Phi} : K[x]/(p(x)) \to K'[x]/(\overline{p}(x)) : f(x) + (p(x)) \mapsto$$

$\square$

0.2 Extensions of Field Isomorphisms

> **Theorem 0.2.1.** Let E be a finite extension of F. Let $\varphi : F \to F'$ be a field isomorphism.
> Then, the number of extensions of φ to E is finite.
> Rigorously, $\{\overline{\varphi} : E \to \overline{F'} \mid \overline{\varphi}|_F = \varphi,\ \overline{\varphi}$ is a field homomorphism$\}$ is finite.

Proof. Since E is finite-dimensional over F, there exist $\alpha_1, \ldots, \alpha_n \in E$ s.t. $E = F(\alpha_1, \ldots, \alpha_n)$.
For each $1 \leq i \leq n$, let

$$\operatorname{irr}(\alpha_i, F) = a_{i,0} + a_{i,1}x + \cdots + a_{i,m_i}x^{m_i} \in F[x]$$

If $\overline{\varphi} : E \to \overline{F'}$ is a homomorphism s.t. $\overline{\varphi}|_F = \varphi$, then $\overline{\varphi}(\alpha_i)$ must be a root of

$$\varphi(a_{i,0}) + \varphi(a_{i,1})x + \cdots + \varphi(a_{i,m_i})x^{m_i} \in F'[x]$$

Because

$$\sum_{k=0}^{m_i} \varphi(a_{i,k}) \cdot (\overline{\varphi}(\alpha_i))^k = \overline{\varphi}\left(\sum_{k=0}^{m_i} a_{i,k} \cdot \alpha_i^k\right) = \overline{\varphi}\left(\operatorname{irr}(\alpha_i, F)(\alpha_i)\right) = \overline{\varphi}(0) = 0$$

Therefore, the number of extensions is at most

$$\deg \operatorname{irr}(\alpha_1, F) \cdot \deg \operatorname{irr}(\alpha_2, F) \cdots \deg \operatorname{irr}(\alpha_n, F).$$

□

Specifically, note that: E/F is a finite extension $\iff$ $E = F(\alpha_1, \ldots, \alpha_n)$ where $\alpha_i \in E$.
Let $\alpha \in \overline{F}$ be algebraic over F. Then there exists a monic irreducible

$$f(x) = a_0 + a_1x + \cdots + a_{n-1}x^{n-1} + x^n \in F[x] \quad (n \geq 1)$$

satisfying $f(\alpha) = 0$. Let $\varphi : F \to F'$ be an isomorphism. We have shown that $\overline{\varphi} : F(\alpha) \to \overline{F'}$ s.t. $\overline{\varphi}|_F = \varphi$ exists.
Since $F(\alpha) = \operatorname{span}_F\{1, \alpha, \ldots, \alpha^{n-1}\}$, for any $\sum_{i=0}^{n-1} a_i\alpha^i \in F(\alpha)$,

$$\overline{\varphi}\left(\sum_{i=0}^{n-1} a_i\alpha^i\right) = \sum_{i=0}^{n-1} \varphi(a_i) \cdot \overline{\varphi}(\alpha)^i \in \overline{\varphi}[F(\alpha)] \leq \overline{F'}$$

That is, $\overline{\varphi}$ is uniquely determined by the value of $\overline{\varphi}(\alpha)$ in $\overline{F'}$.

Theorem 0.2.2. Let E be an algebraic extension of F.
Let $\varphi_1 : F \to F_1$ and $\varphi_2 : F \to F_2$ be field isomorphisms.
Then, there exists a bijection between

$$\begin{aligned} S_1 &= \{\widehat{\varphi}_1 : E \to \overline{F}_1 \mid \widehat{\varphi}_1 \text{ is a field homomorphism, } \widehat{\varphi}_1|_F = \varphi_1\} \\ &\updownarrow \\ S_2 &= \{\widehat{\varphi}_2 : E \to \overline{F}_2 \mid \widehat{\varphi}_2 \text{ is a field homomorphism, } \widehat{\varphi}_2|_F = \varphi_2\} \end{aligned}$$

Proof. Let $\Phi : \overline{F}_1 \to \overline{F}_2$ be an extension isomorphism of $\varphi_2 \circ \varphi_1^{-1}$.
Let $\widehat{\varphi}_1 \in S_1$ be given. Then, a map

$$\Phi \circ \widehat{\varphi}_1 : E \to \overline{F}_2;\ \ a \mapsto \Phi(\widehat{\varphi}_1(a))$$

is a field homomorphism s.t. $(\Phi \circ \widehat{\varphi}_1)|_F = \varphi_2$, because for any $a \in F$,

$$\Phi(\widehat{\varphi}_1(a)) = \Phi(\varphi_1(a)) = (\varphi_2 \circ \varphi_1^{-1})(\varphi_1(a)) = \varphi_2(a).$$

Apply the same process reversely with respect to Φ^{-1}, then we obtain a 1-1 correspondence. $\square$

0.2.1 Useful facts for Field and Galois Theory

Lemma 0.2.3. Let R and S be non-zero rings with identity 1_R and 1_S, respectively.
Let $\varphi : R \to S$ be a non-trivial Ring Homomorphism.

1. If $\varphi(1_R) \neq 1_S$, then $\varphi(1_R)$ is a zero divisor in S.
2. If $\varphi(1_R) = 1_S$, then for each unit $u \in R$, $\varphi(u)^{-1} = \varphi(u^{-1})$.

Proof. Since φ preserves the multiplication,

$$\varphi(1_R) \cdot \varphi(1_R) = \varphi(1_R \cdot 1_R) = \varphi(1_R) \implies \varphi(1_R) \cdot (\varphi(1_R) - 1_S) = 0.$$

Since $\varphi(1_R)$ is not 1_S, thus $\varphi(1_R)$ is a zero divisor.
To show the second assertion, let a unit $u \in R$ be given. Then,

$$\varphi(u) \cdot \varphi(u^{-1}) = \varphi(u \cdot u^{-1}) = \varphi(1_R) = 1_S.$$

□

Corollary 0.2.4. If $\varphi : R \to S$ is a Ring Homomorphism and S is an integral domain, then φ preserves an identity.

Lemma 0.2.5. Every Field Automorphism fixes the prime subfield.

Lemma 0.2.6. Non-trivial field Homomorphism is injective.

Proof. Let $\varphi : F \to F'$ be a field Homomorphism. Since $\ker \varphi \trianglelefteq F$ is an ideal of field F, $\ker \varphi$ is either $\{0\}$ or F. □

Note: the codomain is not necessarily a field, but a ring.